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DictCR-former: Content-aware dictionary transformer for cloud removal

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ABSTRACT

Atmospheric conditions frequently result in variable cloud coverage in remote sensing imagery, posing significant challenges for downstream applications. While deep learning has demonstrated effectiveness in cloud removal, existing single-image approaches may not sufficiently compensate for critical information voids caused by cloud occlusion, making accurate restoration particularly challenging. To address these limitations, we propose DictCR-former, a transformer-based architecture that incorporates dictionary-inspired operators to enable cross-image feature aggregation through our Content-Aware Dictionary Attention (CADict-Att). The CADict-Att synergizes two core components: (1) Content-Aware Dictionary Learning (CADL) that dynamically constructs dictionary atoms by capturing latent cross-image correlations to encode fundamental semantic patterns; and (2) Dictionary Aggregation (DA) which adaptively combines contextually relevant atoms through learnable attention weights, reconstructing cloud-contaminated regions while maintaining spatial consistency with unobscured areas. Extensive experimental evaluations on multiple benchmarks demonstrate that DictCR-former outperforms existing methods in cloud removal, achieving superior performance while maintaining computational efficiency.

1. Introduction

Remote sensing data plays a crucial part in geoscientific research, which enables the continuous monitoring of the Earth's surface and supporting various applications, such as resource exploration, agricultural management, and disaster assessment [1]. However, cloud formations obscure approximately 67% of the Earth's surface and 55% of the land surface [2], significantly impeding the effectiveness of subsequent analyses and applications. Consequently, developing techniques for cloud removal has emerged as a crucial research area.

Cloud formation in remote sensing imagery is influenced by various factors, particularly atmospheric conditions, resulting in diverse cloud structures with varying densities, shapes, and coverage patterns [3]. Despite the inherent complexities associated with cloud removal, recent advancements in deep learning algorithms, such as DSen2-CR [3] and CR-former [4], which utilize convolution or attention to model contextual relationships, have demonstrated exceptional performance. However, these methods primarily establish contextual relationships within distinct regions of a single input image. This approach encounters a significant challenge due to its over-reliance on visual context: cloud interference can obscure certain regions of the image, preventing the extraction of essential information needed for establishing reliable contextual relationships. Consequently, the acquired contextual information often fails to capture critical associations between image regions, resulting in degraded image restoration quality.

In this study, we introduce a novel paradigm, the DictCR-former network, for cloud removal that learns cross-image contextual information. In DictCR-former, we construct a series of dictionary atoms to store features across multiple images. During cloud removal, the network retrieves relevant cross-image contextual features through the proposed Content-Aware Dictionary Attention mechanism (CADict-Att) to identify features potentially pertinent to the target region. Unlike previous methods that rely solely on a single image to learn features, our approach utilizes an external memory bank storing dictionary atoms to provide richer cross-image contextual information. This allows regional information from different images to be propagated to the damaged areas of the target image, thereby achieving improved restoration of the missing content.

The CADict-Att mechanism consists of two key components: Content-Aware Dictionary Learning (CADL) and Dictionary Aggregation (DA). The CADL module dynamically constructs dictionary atoms

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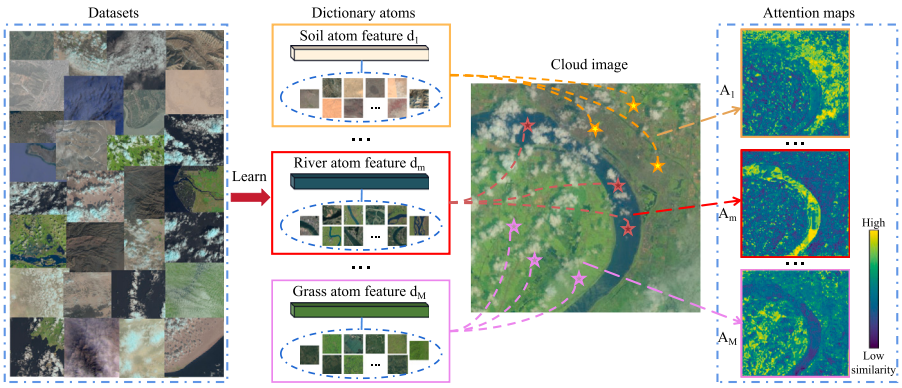


Fig. 1. An illustration of the CADict-Att mechanism. The process begins by learning dictionary atoms from the training dataset, followed by computing attention maps between the learned atoms and cloud-contaminated image features. These attention maps demonstrate a strong semantic alignment between the dictionary atoms and features in cloud-contaminated images, effectively capturing fundamental patterns in cloud-obscured areas.

from the training data via cross-attention, effectively capturing cross-image correlations and encoding fundamental semantic patterns. The DA module adaptively aggregates context-relevant dictionary atoms using learnable attention weights, generating key features for the precise reconstruction of cloud-covered images. By effectively capturing and aggregating cross-image correlations, DictCR-former significantly enhances the accuracy of cloud removal in remote sensing images.

Fig. 1 illustrates the CADict-Att module within the proposed DictCR-former. The process first learns dictionary atoms from the training dataset to capture cross-image correlations across fundamental semantic patterns (e.g., soil, rivers, and grassland). Similarity scores then generate attention maps that correlate these atoms with cloudy image features, guiding the reconstruction of occluded regions. Comprehensive evaluations across the RICE1, RICE2 [5], T-CLOUD [6], and SEN12MS-CR [3] datasets consistently demonstrate the model's superior performance.

The key contributions of this study are summarized as follows:

- We introduce DictCR-former, an advanced dictionary-based transformer architecture that leverages cross-image contextual features for high-fidelity cloud removal.
- We propose the Content-Aware Dictionary Attention (CADict-Att) mechanism, which dynamically constructs dictionary atoms and aggregates contextually relevant features using learnable cross-attention weights to restore cloud-obscured regions.
- Extensive experiments demonstrate that DictCR-former outperforms state-of-the-art methods in reconstruction accuracy while maintaining strong spatial consistency.

The remainder of this paper is structured as follows. Section 2 reviews related work. Section 3 details the DictCR-former architecture. Section 4 presents experimental results and comparisons, and Section 5 concludes the paper.

2. Related works

2.1. Traditional cloud removal methods

Before the dominance of deep learning, cloud removal relied on physical models and signal processing. These traditional approaches included multi-temporal pixel replacement [7], multi-source optical-SAR fusion [3], single-image spatial inpainting [8], and spectral-domain filtering utilizing inter-band correlations [9].

To leverage multiple information streams, hybrid strategies integrated spatial, spectral, and temporal domains [10]. Notably, sparse representation with dictionary learning (e.g., via basis pursuit, orthogonal matching pursuit [11], or K-SVD [12]) emerged as a sophisticated hybrid approach. These methods reconstruct obscured regions by modeling image patches as sparse linear combinations of atoms learned from clear data. While effective, traditional dictionary methods were heavily constrained by the limited expressive capacity of linear models, high computational costs, and a strict reliance on hand-crafted priors, which ultimately motivated the shift toward data-driven deep learning solutions.

2.2. Deep learning-based cloud removal methods

Deep learning enables end-to-end learning of complex nonlinear mappings for remote sensing cloud removal [13]. Existing methods can be categorized by their input data (single-image vs. auxiliary) and prior-modeling strategies (local/global features vs. cross-image dictionary priors). This section reviews single-image, auxiliary data-based, and deep dictionary learning methods, identifying their respective limitations to motivate our proposed framework.

2.2.1. Single-image cloud removal

Single-image methods restore cloud-obscured regions using only the information within the target image. Their evolution reflects a shift from local feature modeling (via CNNs/GANs) to global contextual modeling (via Transformers and Diffusion models) to better handle large-scale or broken cloud cover.

Early CNN- and GAN-based approaches, such as RSC-Net [14] and Cloud-GAN [15], extract hierarchical local features or enhance realism through adversarial training. However, their fixed local receptive fields limit their efficacy under large-scale cloud cover, where coherent restoration requires long-range dependencies.

To address this, Transformer-based and attention-augmented methods model global relationships via self-attention [16]. Key works include DecloudFormer [17], efficient linear attention mechanisms for high-resolution imagery [4], and specialized modules for dehazing [18] or cloud density estimation [19]. Hybrid models like SPAGAN [20] further integrate attention into CNN backbones. Although successful, these methods focus primarily on intra-image relationships and do not explicitly learn a compact set of reusable, cross-image priors.

Recently, diffusion models have emerged to generate realistic outputs via iterative denoising. Representative frameworks include DE-MemoryNet [21] for texture refinement, DiffCR [22] for accelerated inference, SeqDMs [23] for multi-modal inputs, and DLFN [24] for frequency-aware enhancement. To mitigate structural distortions, recent work like LSM [25] explicitly models latent structural topology to reinforce semantic consistency. Despite their generative merits, diffusion models suffer from artifacts (e.g., pseudo-textures) and high computational overhead, which limits their practical deployment compared to feedforward architectures.

2.2.2. Auxiliary data-based cloud removal

Auxiliary data-based methods leverage external information—either multi-temporal (same location, different times) or multi-source (different sensors)—to supplement single-image context when internal information is insufficient.

Multi-temporal methods exploit the temporal continuity of land surfaces. For instance, a spatio-temporal neural network [26] fuses multi-temporal Landsat data via 2D convolutions to reconstruct thick clouds, while a 3D CNN [27] models dynamics across optical/radar sequences. However, sudden surface changes (e.g., wildfires) easily invalidate these continuity assumptions.

Multi-source methods fuse complementary sensor data to bypass cloud interference. Optical-SAR fusion frameworks, such as DSen2-CR [3] and GLGF-CR [28], align SAR-derived structural priors with optical spectral details. Similarly, cross-platform methods (e.g., Sentinel-2 and Sentinel-3 [29]) transfer references to balance spatial and temporal resolutions, while GHRN [30] aligns low-quality auxiliary data for thin cloud removal. To advance this paradigm, recent work integrates global–local feature interactions with conditional consistency constraints into diffusion models [31] for high-precision SAR-guided reconstruction. Despite their accuracy, these methods remain constrained by strict data availability (e.g., satellite revisit cycles) and heavy preprocessing complexity (e.g., geometric registration).

2.2.3. Deep dictionary learning methods

Deep dictionary learning methods focus on cross-image priors, capturing universal atomic patterns (e.g., land textures) from cloud-free datasets to guide restoration without relying on auxiliary data.

By replacing hand-crafted atoms with learnable convolutional filters, these methods enable end-to-end, shift-invariant feature modeling. For instance, CDL [32] uses local sparsity constraints to capture spatial hierarchies, while DCDicL [33] introduces dynamic convolutional atoms to handle spatial variations. Similarly, DD-DCSR [34] utilizes a frequency-division dual-dictionary to separate global structures from fine details. Furthermore, recent frameworks explore dynamic-static hybrid dictionary learning [35] within unrolled networks (e.g., deep K-SVD) to balance structural retention with fine texture restoration.

However, a critical limitation persists: these methods optimize a static dictionary during training, reducing their adaptability to diverse remote sensing landscapes (e.g., deserts vs. rainforests) and often leading to generic reconstructions.

In summary, current paradigms present a clear trade-off: Transformers excel at dynamic intra-image modeling but lack cross-sample priors; deep dictionary learning captures these priors but relies on static dictionaries; and diffusion models suffer from generative artifacts and high computational costs. To break this bottleneck, we propose DictCR-former, which integrates a dynamic, content-aware dictionary within an efficient Transformer framework. This synergistically combines adaptive intra-image context with robust cross-sample priors for high-fidelity cloud removal.

3. Methodology

3.1. Preliminaries

This subsection reviews dictionary learning and attention mechanisms, followed by the motivation for our dictionary-based attention approach to cloud removal.

3.1.1. Dictionary learning

Previous dictionary learning-based methods for cloud removal [12] primarily employed sparse representation techniques to learn a dictionary of basic elements, or "atoms". By finding the sparse coefficients for a given cloudy image, these methods could reconstruct the cloud-free version using the learned dictionary.

In this framework, an image matrix $Y \in \mathbb{R}^{d\times N}$ is approximated as a linear combination of dictionary atoms, formulated as $Y \approx DZ$. Here, $D \in \mathbb{R}^{d\times M}$ is the dictionary matrix where each column is an atom, and $Z \in \mathbb{R}^{M\times N}$ is the matrix of sparse representation coefficients. The variable d denotes the feature dimensionality, N is the number of pixels, and M is the number of dictionary atoms.

For image restoration tasks, this is typically formulated as the bi-level optimization problem shown in Eq. (1). The goal is to learn a dictionary D from a training dataset containing T sample pairs, denoted as $\{Y_i, Y_i^{gt}\}_{i=1}^{T}$, where $Y_i \in \mathbb{R}^{d\times N}$ is the cloudy image and $Y_i^{gt} \in \mathbb{R}^{d\times N}$ is its corresponding cloud-free ground truth.

$$\min_{D} \frac{1}{T}\sum_{i=1}^{T} L\left(DZ_i,\ Y_i^{gt}\right), \quad s.t.\ Z_i = \operatorname{argmin}_Z \left|DZ - Y_i\right|_2^2 + \lambda\psi(Z). \tag{1}$$

The outer objective of this optimization seeks to minimize the average reconstruction error over the training set. The function $L(\cdot,\cdot)$ represents the reconstruction loss, which quantifies the discrepancy between the reconstructed image DZ_i and the ground truth image Y_i^{gt}. Common choices for this function in image restoration include pixel-wise metrics such as the L_2 norm (Mean Squared Error) or the L_1 norm (Mean Absolute Error).

The inner objective (the s.t. constraint) solves for the optimal sparse coefficients Z_i for a given cloudy image Y_i and the current dictionary D. The function $\psi(\cdot)$ is a sparsity-inducing regularizer, typically the l_0 or l_1 norm, and the hyperparameter λ controls the trade-off between reconstruction fidelity and sparsity.

For a given cloudy image Y_i, this approach would first learn the dictionary D from the entire training data, enabling it to encode a rich basis of recurring, dataset-level patterns, often using an iterative method like K-SVD [36]. Then, it would solve for the coefficients Z_i and reconstruct the final cloud-free image as $Y_i' = DZ_i$.

3.1.2. Attention mechanism

The standard attention mechanism [4] dynamically reweights input features to capture long-range dependencies. For an input feature map $X \in \mathbb{R}^{N\times d}$ (N = sequence length, d = feature dimension), its operation is formulated as:

$$Q = XW_Q, K = XW_K, V = XW_V,\quad A = \mathrm{Sim}(Q,K),\quad X_{out} = AV, \tag{2}$$

where W_Q, W_K, W_V are learnable projections, $\mathrm{Sim}(\cdot,\cdot)$ (typically Softmax) computes query-key similarity to generate the attention matrix A, the output X_{out} is a weighted aggregation of the value matrix V.

A critical limitation is that V—acting as a dynamic, instance-specific dictionary—is derived solely from X, confining the mechanism to intra-sample context and precluding cross-sample relationship modeling.

3.1.3. Motivation

Comparing the formulations of dictionary learning (Eq. (1)) and self-attention (Eq. (2)) reveals a powerful analogy. The value matrix V in attention is analogous to the dictionary D, while the attention matrix A acts as the reconstruction coefficients Z. Both methods produce an output by computing a weighted sum of basis vectors.

However, this analogy exposes the fundamental limitation of self-attention that motivates our work. In dictionary learning, the dictionary D is a persistent, rich representation of patterns learned from thousands of training samples. In contrast, the "dictionary" V in self-attention is transient, constructed from and limited to the features of a single input instance. This prevents standard attention mechanisms from explicitly modeling the relationships between different samples.

Therefore, we propose to integrate the key strength of dictionary learning—its ability to build a shared, dataset-level basis of patterns—into the flexible, dynamic framework of attention. This synthesis aims to overcome the single-instance constraint of attention, enabling the model to leverage cross-sample context for more robust feature representation in cloud removal.

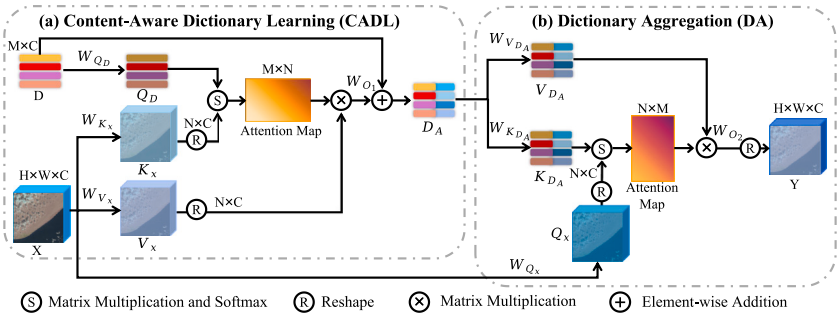


Fig. 2. Illustration of CADict-att mechanism. It comprises two primary components: (a) Content-Aware Dictionary Learning (CADL) and (b) Dictionary Aggregation (DA).

3.2. Content-Aware Dictionary Attention

This subsection introduces the Content-Aware Dictionary Attention (CADict-Att), designed to capture cross-sample correlations for reconstruction. As shown in Fig. 2, CADict-Att consists of two components: Content-Aware Dictionary Learning (CADL) and Dictionary Aggregation (DA).

3.2.1. Content-Aware Dictionary Learning (CADL)

We define a learnable dictionary $D \in \mathbb{R}^{M\times C}$, initialized as network parameters, where M is the number of atoms in the dictionary, and C is the feature dimension. During training, the dictionary atoms D are optimized to capture semantically rich and contextually relevant patterns derived from the entire dataset.

Given an input image feature $X \in \mathbb{R}^{N\times C}$, where $N = H \times W$ corresponds to the resolution of the input image, the dictionary D and the input feature X undergo specific transformations. Specifically, the dictionary D is projected into a query space Q_D using a projection matrix W_{Q_D}. Similarly, the input feature X is projected into the key and value spaces, denoted as K_X and $V_X \in \mathbb{R}^{N\times C}$, using projection matrices W_{K_X} and W_{V_X}, respectively.

Then cross-attention is applied between the dictionary and the input feature to generate content-based dictionary atoms $D_A \in \mathbb{R}^{M\times C}$, as illustrated in Fig. 2(a). The process can be formulated as follows:

$$\begin{aligned} &Q_D = DW_{Q_D}, K_X = XW_{K_X}, V_X = XW_{V_X}, \\ &A_D = \mathrm{softmax}\left(Q_D K_X^T/\sqrt{C}\right), \\ &D'_A = A_D V_X, \\ &D_A = D + D'_A W_{O_1}, \end{aligned} \tag{3}$$

where Softmax function is applied to compute the dictionary-feature attention map $A_D \in \mathbb{R}^{M\times N}$. This attention map quantifies the similarity between the query dictionary Q_D and the key image feature K_X. The adjusted residual atoms D'_A are derived by applying A_D to the value matrix V_X, which captures relevant input features. Subsequently, the final content-aware dictionary atoms D_A are obtained by adding the projected residual component D'_A to the original dictionary D. To ensure computational efficiency, the number of dictionary atoms M is set to be significantly smaller than the number of input features N ($M \ll N$).

This CADL component effectively leverages the dictiona-ry-feature attention map A_D to extract relevant input features from V_X, resulting in the refinement of dictionary atoms. This process is analogous to dictionary learning, as outlined in Eq. (1), where dictionary atoms are iteratively optimized based on training data.

3.2.2. Dictionary Aggregation (DA)

After constructing the content-aware dictionary from the training samples, another cross-attention operation is used to capture correlations between the dictionary and the input image features. These correlations adjust the weights of the dictionary's fundamental atoms, facilitating the recovery of features essential for reconstructing missing or cloud-obscured regions.

The process begins with projecting the input feature X into the query space Q_X using the projection matrix W_{Q_X}. Simultaneously, the content-aware dictionary D_A is projected into the key and value spaces, denoted by K_{D_A} and V_{D_A}, using projection matrices $W_{K_{D_A}}$ and $W_{V_{D_A}}$, respectively. The cross-attention operation, illustrated in Fig. 2(b), is formulated as follows:

$$\begin{aligned} &Q_X = XW_{Q_X}, K_{D_A} = D_A W_{K_{D_A}}, V_{D_A} = D_A W_{V_{D_A}}, \\ &A_F = \mathrm{softmax}\left(Q_X K_{D_A}^T/\sqrt{C}\right), \\ &Y' = A_F V_{D_A}, \\ &Y = Y'W_{O_2}, \end{aligned} \tag{4}$$

where $A_F \in \mathbb{R}^{N\times M}$ represents the attention map, which captures the similarity between the query features Q_X and the key dictionary elements K_{D_A}. This attention map weights the value dictionary elements V_{D_A} to produce the feature Y'. Subsequently, Y' undergoes transformation using the projection matrix W_{O_2} to produce the final feature representation Y.

The DA component computes the feature-dictionary attention map A_F, analogous to the representation coefficients in dictionary learning (Eq. (1)). These attention values aggregate the corresponding dictionary elements to propagate enhanced features, similar to the reconstruction of cloud-free images using dictionary atoms and their associated coefficients.

By integrating the Content-Aware Dictionary Learning (CADL) and Dictionary Aggregation (DA) components, the proposed CADict-Att mechanism adaptively incorporates cross-image relationships into the learned dictionary atoms and combines contextually relevant atoms to generate refined image features. This methodology facilitates efficient and high-performance restoration of cloud-covered regions in imagery.

3.2.3. Complexity analysis

The number of dictionary atoms M is significantly smaller than the resolution of the input image feature N. Consequently, the attention maps A_D and A_F produced by the proposed method are considerably smaller than the original attention map A defined in Eq. (2). The computational complexity and memory requirements of the conventional attention mechanism are $\mathcal{O}(N^2C)$ and $\mathcal{O}(N^2)$, respectively. In contrast, the CADict-Att mechanism effectively reduces the computational complexity to $\mathcal{O}(NMC)$ and the memory cost to $\mathcal{O}(NM)$, where $M \ll N$. This optimization reduces the attention mechanism's complexity from quadratic to linear with respect to the input feature length N, improving computational efficiency.

3.3. Architecture of DictCR-former

To facilitate the proposed attention mechanism, we construct the DictCR-former using a standard hierarchical framework. As the primary

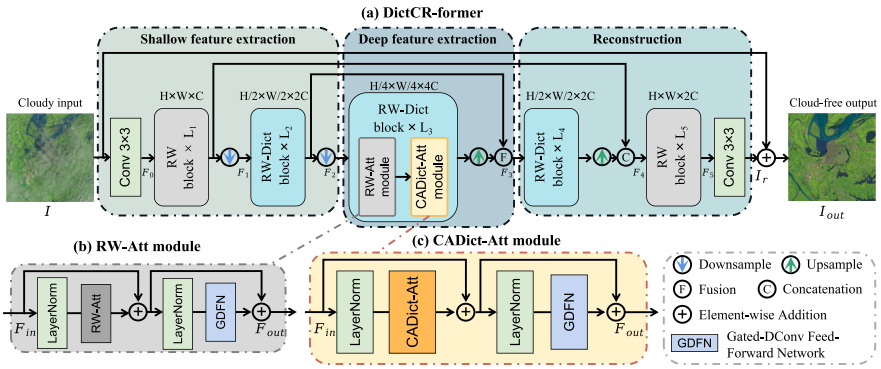


Fig. 3. Architecture of the DictCR-former. (a) The overall hierarchical U-Net-style framework. (b) The RW-Att module designed for modeling local regional dependencies. (c) The proposed CADict-Att module designed to capture global cross-image patterns via content-aware dictionary attention..

contribution of this work lies in the CADict-Att module, we adopt a widely established U-Net-style architecture for the overall network, as illustrated in Fig. 3(a).

3.3.1. Network architecture

The DictCR-former is built upon a hierarchical encoder–decoder framework, organized into three distinct stages: shallow feature extraction, deep feature extraction, and reconstruction. Initially, a convolutional layer projects the input cloudy image I into a high-dimensional shallow feature space. These features subsequently traverse a symmetric U-Net structure composed of cascaded Transformer blocks. The encoder pathway progressively reduces spatial resolution via downsampling to expand the receptive field, while the decoder pathway recovers spatial details through corresponding upsampling operations and skip connections. Finally, the reconstructed features are fused with the original input via a global residual connection to yield the cloud-free output I_{out}. This streamlined architecture allows the model to prioritize the learning of robust feature representations while minimizing unnecessary computational overhead.

3.3.2. Transformer block

The fundamental building blocks of the network are the Transformer layers, which are strategically configured to handle different scales of information. We employ a hybrid attention strategy to effectively balance local spatial processing with global dictionary-based reasoning.

Specifically, the initial and final transformer blocks utilize RW blocks equipped with the RW-Att module [37], as shown in Fig. 3(b). These blocks focus on capturing short-range regional dependencies efficiently. Conversely, the intermediate layers employ RW-Dict blocks, which alternate between the RW-Att and the proposed CADict-Att modules. As illustrated in Fig. 3(c), the integration of CADict-Att allows the network to explicitly model cross-image correlations through the content-aware dictionary mechanism, thereby enriching the semantic context in the deep feature space.

Structurally, both block variants adhere to a standard Transformer formulation, comprising two LayerNorm operations [38], the respective attention mechanism, and a Gated-DConv Feed-Forward Network (GDFN) [39]. Formally, the computational flow within a block, given an input feature F_{in}, is defined as:

$$\begin{aligned} F' &= F_{in} + \mathrm{Att}\left(\mathrm{LayerNorm}(F_{in})\right), \\ F_{out} &= F' + \mathrm{GDFN}\left(\mathrm{LayerNorm}(F')\right), \end{aligned} \tag{5}$$

where Att denotes either the RW-Att or CADict-Att operation, F' represents the intermediate features, and F_{out} denotes the block output. This architectural design enables DictCR-former to synergistically capture local texture details and global structural patterns, ensuring robust restoration performance across diverse cloud removal scenarios.

4. Experiments

This section details the experimental setup, evaluates DictCR-former against state-of-the-art baselines, and validates the CADict-Att mechanism via ablation studies.

4.1. Experimental settings

4.1.1. Datasets and evaluation metrics

The DictCR-former method is evaluated on four widely used cloud removal datasets: RICE1 [5], RICE2 [5], T-CLOUD [6], and SEN12MS-CR [3].

- RICE1 [5]: The synthetic thin-cloud dataset comprises 500 image pairs sourced from Google Earth, each with a resolution of 512 × 512 pixels. A randomly selected subset of 100 image pairs is used for testing.
- RICE2 [5]: The thick-cloud dataset contains 735 pairs of natural color images derived from Landsat 8, each with a resolution of 512 × 512 pixels. Cloud-free images were manually selected from the same locations and captured within 15 days of the cloud-covered counterparts. A randomly selected subset of 147 image pairs is used for testing.
- T-CLOUD [6]: The real-world thin-cloud dataset comprises 2939 pairs based on Landsat 8 RGB images, with a resolution of 256 × 256 pixels. The image pairs are acquired under comparable lighting conditions by the same satellite at different times. A predefined test set of 588 image pairs is used.
- SEN12MS-CR [3]: The large-scale multispectral dataset encompasses diverse global regions across all seasons. It comprises triplets of 13-band Sentinel-2 optical images (cloud-covered and cloud-free pairs) and 2-band Sentinel-1 SAR images. The dataset includes 175 regions of interest (ROIs), partitioned into 256 × 256 pixel patches, with approximately 48% cloud coverage. The dataset contains 114,319 triplets for training and 7899 triplets for testing.

Our method's performance is evaluated using four widely adopted metrics: mean absolute error (MAE), peak signal-to-noise ratio (PSNR), structural similarity index (SSIM) and spectral angle mapper (SAM).

4.1.2. Implementation details

The DictCR-former model is implemented using the PyTorch framework and trained on an NVIDIA A800 GPU with 80 GB of memory. The optimization process uses the AdamW optimizer [44] with an L_1 loss function. To ensure a fair comparison across all experiments, the batch size was consistently set to 24, and the model was trained on cropped 256 × 256 image patches. The training duration and learning rate schedules were carefully tailored to the scale and complexity of

Table 1
Quantitative comparison results on RICE1 and RICE2. Metrics with ↓ indicate lower is better, and ↑ indicate higher is better. The best results are in bold.

Models	RICE1				RICE2			
	MAE↓	SAM↓	PSNR↑	SSIM↑	MAE↓	SAM↓	PSNR↑	SSIM↑
Pix2Pix [40]	0.0253	4.11	31.97	0.9161	0.0327	5.08	29.45	0.8473
SPA-GAN [20]	0.0372	2.25	28.89	0.9144	0.0403	3.36	27.51	0.8177
CVAE [6]	0.0198	1.11	33.70	0.9562	0.0216	1.69	33.62	0.9079
Restormer [39]	0.0176	1.01	35.42	0.9617	0.0163	1.27	36.05	0.9155
CMNet [41]	0.0144	0.98	36.26	0.9625	0.0167	1.29	35.82	0.9151
ACA-CRNet [42]	0.0151	0.98	36.05	0.9628	0.0164	1.29	35.65	0.9126
CR-former-L [4]	0.0138	0.92	36.75	0.9627	0.0163	1.26	36.24	0.9161
CR-Famba [43]	0.0325	1.64	30.91	0.9142	0.0252	2.23	31.90	0.8495
DE-MemoryNet [21]	0.0169	1.21	34.96	**0.9721**	0.0183	1.59	34.43	0.8982
Ours	**0.0132**	**0.89**	**37.14**	0.9630	**0.0148**	**1.15**	**36.76**	**0.9189**

Table 2
Quantitative comparison results on T-CLOUD. Metrics with ↓ indicate lower is better, and ↑ indicate higher is better. The best results are in bold.

Models	T-CLOUD			
	MAE↓	SAM↓	PSNR↑	SSIM↑
Pix2Pix [40]	0.0449	10.01	25.64	0.7563
SPA-GAN [20]	0.0419	4.09	26.14	0.7954
CVAE [6]	0.0342	2.95	28.19	0.8613
Restormer [39]	0.0248	2.46	30.49	0.8851
CMNet [41]	0.0251	2.49	30.33	0.8829
ACA-CRNet [42]	**0.0234**	2.35	30.85	0.8885
CR-former-L [4]	0.0238	2.43	30.82	0.8867
CR-Famba [43]	0.0262	2.48	30.10	0.8810
Ours	0.0237	**2.34**	**30.88**	**0.8905**

Table 3
Quantitative comparison results on SEN12MS-CR. Metrics with ↓ indicate lower is better, and ↑ indicate higher is better. The best results are in bold.

Models	SEN12MS-CR			
	MAE↓	SAM↓	PSNR↑	SSIM↑
Pix2Pix [40]	0.031	10.784	27.60	0.864
SPA-GAN [20]	0.045	18.085	24.78	0.754
DSen2-CR [3]	0.031	9.472	27.76	0.874
GLF-CR [46]	0.028	8.981	28.64	0.885
UnCRtainTS [47]	0.027	8.320	28.90	0.880
ACA-CRNet [42]	0.025	7.770	29.78	0.896
Ours	**0.024**	**7.664**	**29.86**	**0.901**

each dataset, aligning with established practices [3,43] in the field. For the small-to-medium scale datasets (RICE1, RICE2, and T-CLOUD), the model was trained for 1000 epochs to ensure full convergence. On RICE1 and RICE2, the learning rate was initialized to 4×10^{-4} and decayed to 2×10^{-6} using a cosine annealing schedule [45]. On T-CLOUD, the learning rate started at 8×10^{-4} and was halved at epochs 460 and 920. For the large-scale, multi-modal SEN12MS-CR dataset, training was conducted for 30 epochs, a duration sufficient for convergence given the large data volume. The learning rate was initialized to 6×10^{-4} and halved every 7 epochs after the initial 10 epochs to accommodate the dataset's complexity. The network architecture features a base channel size of 48 and employs transformer blocks arranged in the sequence $[1, 2, 8, 2, 1]$. The RW-Att module utilizes $[8, 32]$ split sizes, while the CADict-Att module leverages a dictionary comprising $M = 128$ atoms. The source code and pre-trained models will be publicly available on GitHub.

4.2. Performance of DictCR-former

Quantitative and qualitative evaluations across four datasets (Tables 1, 2, 3 and Figs. 4, 5, 6) demonstrate DictCR-former's superiority over SOTA methods.

4.2.1. Quantitative comparison with baselines

DictCR-former was compared with several attention-based methods, including SPA-GAN [20], CVAE [6], Restormer [39], CMNet [41], ACA-CRNet [42], CR-former [4], and GLF-CR [46], CNN-based models such as Pix2Pix [40] and DSen2-CR [3], as well as other methods such as CR-Famba [43] and DE-MemoryNet [21].

The quantitative results are presented in Tables 1–3. On the RICE1, RICE2, and T-CLOUD datasets (Tables 1 and 2), DictCR-former demonstrated superior performance, achieving average improvements of 5.03% in MAE, 5.59% in SAM, 0.90% in PSNR, and 0.26% in SSIM over CR-former, the previous SOTA method. Similarly, on the SEN12MS-CR dataset (Table 3), DictCR-former surpassed ACA-CRNet, achieving improvements of 4.17% in MAE, 1.40% in SAM, 0.27% in PSNR, and 0.56% in SSIM.

The computational efficiency of our DictCR-former is shown in Table 4. It balances model size and computational cost effectively: with only 5.93M parameters, it is nearly 3x lighter than CMNet (16.51M) and 9x lighter than Pix2Pix (54.41M). At 50.8 GFLOPs, it requires over 24x less computation than heavyweights like DSen2-CR (1240.2 GFLOPs) and ACA-CRNet (1422.0 GFLOPs). This efficiency is achieved with competitive GPU memory and practical inference speed, highlighting strong performance without high computational costs.

These findings underscore the superior performance and efficiency of DictCR-former, positioning it as an effective approach for cloud removal.

4.2.2. Qualitative comparison with baselines

Figs. 4–6 present visual comparisons across the RICE, T-CLOUD, and SEN12MS-CR datasets. Our method consistently outperforms baselines in image integrity and consistency across diverse scenes (e.g., islands, urban areas, and rivers).

On RICE and T-CLOUD, CNN-based models (Pix2Pix, SPA-GAN) exhibit grid artifacts and color distortions, while attention-based methods (Restormer, CMNet, ACA-CRNet) struggle with dense clouds due to limited receptive fields. Although CR-former captures global dependencies, it fails to reconstruct discontinuous features (e.g., rivers) as it relies solely on intra-image attention. In contrast, our method effectively restores detailed structures by leveraging the CADict-Att module, which models content-aware cross-image patterns to fill occluded regions. For the multispectral SEN12MS-CR dataset (Fig. 6), baselines like DSen2-CR and GLF-CR suffer from severe color distortion and blurring. Conversely, our approach achieves exceptional clarity and texture recovery. These results validate that utilizing cross-image priors from the entire dataset enables high-quality reconstruction even in heavily clouded multispectral data.

4.3. Ablation studies

We conduct ablation studies on four aspects: CADict-Att effectiveness, attention map visualization, other attention mechanisms, and the impact of hyperparameters (transformer blocks and M).

Table 4
Computational overhead comparison with baseline methods. Metrics include parameters (M), floating-point operations (G), inference time (ms) on an NVIDIA 3090 GPU, and GPU memory consumption (MB).

Models	Params(M)	FLOPs(G)	Inference time (ms)	GPU memory (MB)
Pix2Pix [40]	54.41	6.1	4.6	3357
SPA-GAN [20]	0.21	15.2	26.0	3749
CVAE [6]	15.42	37.1	15.1	1097
Restormer [39]	26.13	155.0	112.3	1629
CMNet [41]	16.51	236.0	197.0	1607
DSen2-CR [3]	18.92	1240.2	161.2	6230
ACA-CRNet [42]	20.39	1422.0	223.7	11 000
CR-former-L [4]	26.15	155.0	109.0	3461
CR-Famba [43]	174.94	165.0	119.5	1716
DE-MemoryNet [21]	36.80	397.7	124.6	13 112
Ours	5.93	50.8	64.9	1531

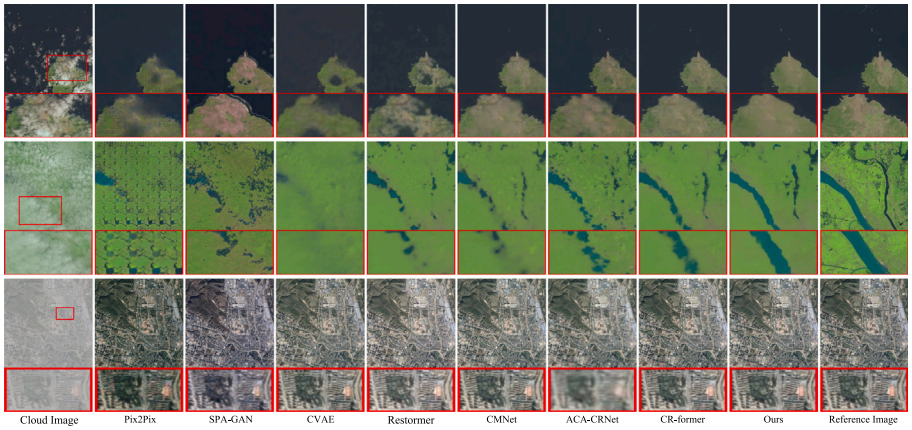


Fig. 4. Visualization of cloud removal results for the RICE1 and RICE2 datasets. Local details (red boxes) are enlarged at the bottom for clarity. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

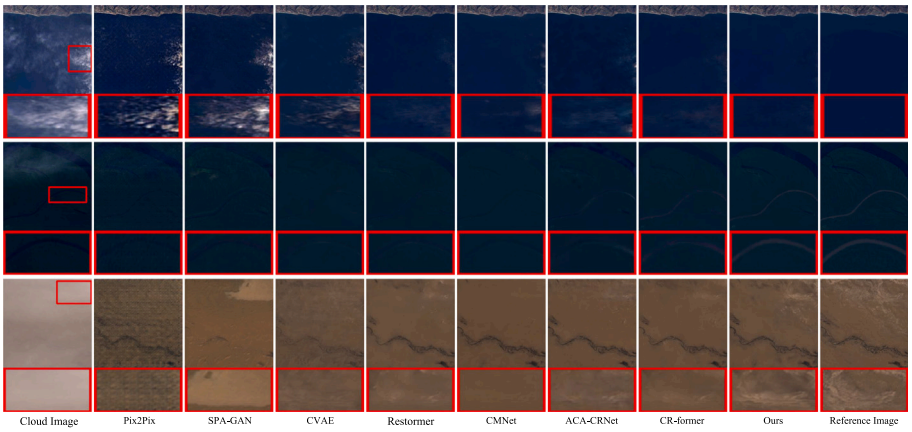


Fig. 5. Visualization of cloud removal results for the T-CLOUD dataset. Local details (red boxes) are enlarged at the bottom for clarity. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.3.1. Effectiveness of CADict-Att

We evaluate the CADict-Att module via four variants: (1) Backbone: A baseline network utilizing the standard RW-Att module [37]; (2) w CADL: Integrates only the CADL module, applying the learned dictionary via direct matrix multiplication while omitting the DA step; (3) w DA: Employs only the DA module with a predefined dictionary in place of content-aware atoms; (4) DictCR-former (w CADL+DA): Our full proposed model, where CADict-Att modules (combining both CADL and DA) replace the RW-Att modules.

Analysis of the quantitative results, presented in Table 5, reveals two key insights. First, our full model, DictCR-former, consistently and significantly outperforms the backbone and both single-component variants across all metrics. Second, the performance of the w CADL and w DA models is severely degraded compared to the full model. For instance, on the RICE1 dataset, DictCR-former achieves the highest PSNR of 37.14, whereas the w CADL model drops to 36.18 and the w DA model only reaches 37.01. This performance gap underscores that neither learning a dictionary nor performing aggregation with a generic one is sufficient on its own. It is the powerful synergy—where

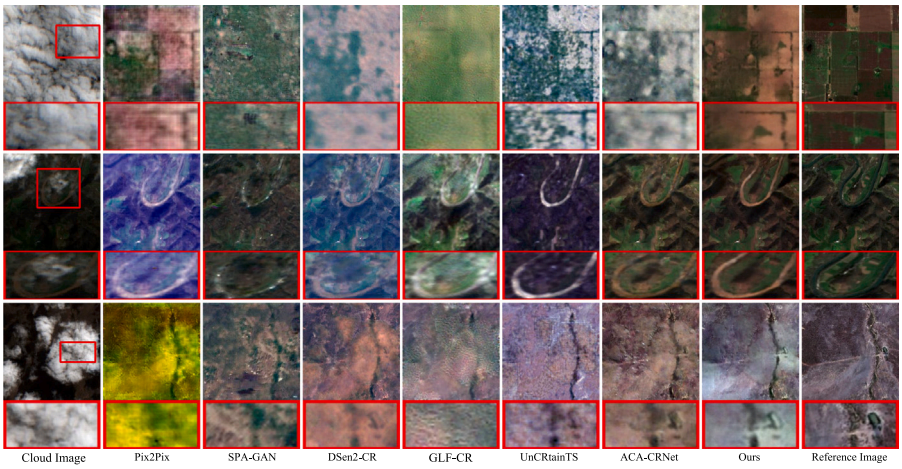


Fig. 6. Visualization of cloud removal results for the SEN12MS-CR dataset. Local details (red boxes) are enlarged at the bottom for clarity. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

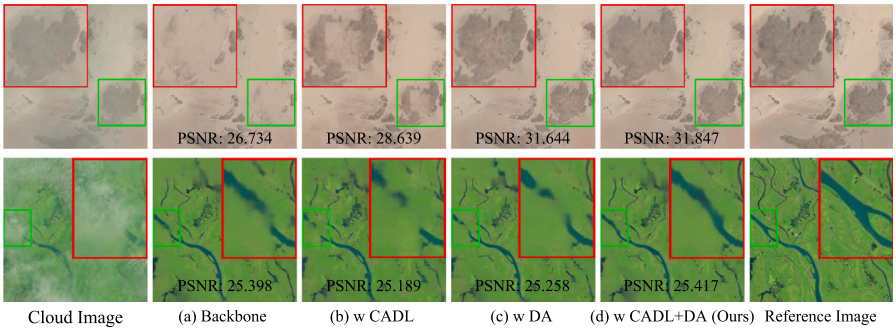


Fig. 7. Visualization of CADict-Att's effectiveness on the RICE1 and RICE2 datasets. (a) Backbone, (b) w CADL, (c) w DA, (d) w CADL+DA (Our DictCR-former). Local details in green boxes are enlarged within red boxes for better clarity. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 5
Effectiveness of CADict-Att. Metrics with ↓ indicate lower is better, and ↑ indicate higher is better. The best results are in bold.

Datasets	Models	MAE↓	SAM↓	PSNR↑	SSIM↑
RICE1	Backbone	0.0136	0.94	36.91	0.9625
	w CADL	0.0149	0.98	36.18	0.9608
	w DA	0.0136	0.92	37.01	0.9627
	w CADL+DA (Ours)	**0.0132**	**0.89**	**37.14**	**0.9630**
RICE2	Backbone	0.0150	1.16	36.56	0.9187
	w CADL	0.0156	1.25	36.24	0.9144
	w DA	0.0154	1.17	36.33	0.9181
	w CADL+DA (Ours)	**0.0148**	**1.15**	**36.76**	**0.9189**

CADL provides the essential content-aware patterns and DA effectively leverages them—that drives the high-fidelity image restoration.

The qualitative improvements are visually illustrated in Fig. 7. The backbone model and the single-component variants exhibit clear artifacts, failing to reconstruct fine-grained soil textures and maintain the structural continuity of the river. In contrast, our full model successfully reconstructs these details, demonstrating the effectiveness of the CADict-Att module in preserving object integrity and spatial consistency.

Despite these improvements, extreme scenarios reveal certain boundary limitations. As shown in the second row of Fig. 7, when heavy cloud cover leads to absolute signal attenuation and leaves complex topologies (e.g., narrow river branches) devoid of contextual clues, the single-component variants (b) and (c) exhibit more missing parts than the baseline (a). Furthermore, our full model (d) omits a fine river branch compared to the Reference Image, indicating that the CADict-Att mechanism tends to prioritize global texture coherence over localized topological accuracy under severe occlusion. This constraint highlights the necessity for dynamic online dictionary learning to improve regional adaptation, as well as the integration of multi-modal data (specifically SAR) to introduce physical constraints for better structural preservation.

Comprehensive evaluations confirm that CADict-Att's integration of dictionary learning and adaptive aggregation allows DictCR-former to capture cross-image patterns, ensuring accurate and spatially consistent cloud-free restoration.

4.3.2. *Analysis of CADict-Att attention maps*

The CADict-Att mechanism dynamically constructs dictionary atoms by capturing latent cross-image correlations, with each atom encoding semantically meaningful patterns. Its effectiveness—critical to DictCR-former's performance—is validated through qualitative and quantitative analyses.

Qualitatively, attention map visualizations (Fig. 8) show that in cloud-free scenes, high similarity scores cluster in distinct regions (grasslands, rivers, etc.), each mapping to specific atoms. In cloud-covered scenes, the same atoms are activated for underlying structures (e.g., cloud-obscured rivers), proving the dictionary penetrates cloud interference to identify invariant features.

Quantitatively, we analyzed top-1 attention scores (the most similar atom for each pixel) and their spatial distribution (Fig. 9). Features from distinct classes cluster tightly around specific atoms: ocean around Atom 34, grassland near Atom 53, and clouds around Atom 110.

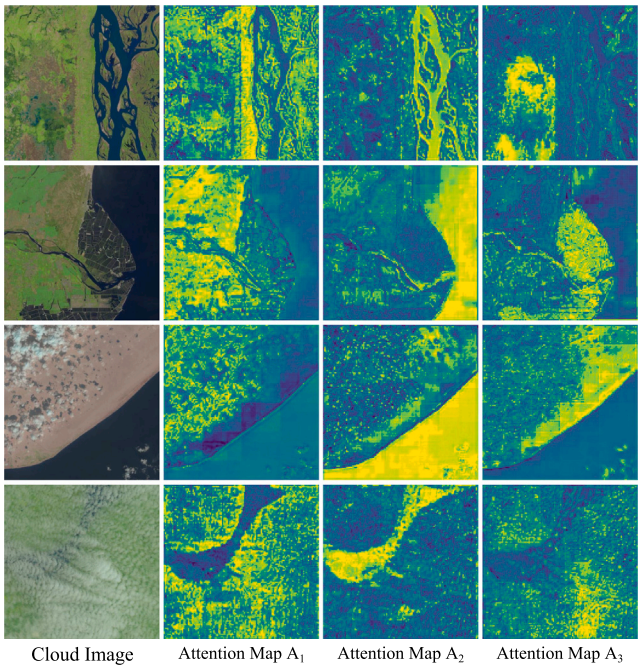


Fig. 8. Visualization of CADict-Att attention maps: Validating semantic consistency of dictionary atoms. High similarity scores cluster in semantically distinct regions, with each region mapping to a specific dictionary atom. Rows 1–2 (cloud-free scenes) and Rows 3–4 (cloud-covered scenes) show atoms consistently focusing on meaningful surface regions (e.g., grasslands, rivers), confirming that dictionary atoms capture invariant scene features.

Table 6
Comparison with different attention mechanisms. The CADict-Att module in our framework was replaced with other SOTA attention variants. Metrics with ↓ indicate lower is better, and ↑ indicate higher is better. The best results are in bold.

Datasets	Methods	MAE↓	SAM↓	PSNR↑	SSIM↑
RICE1	RW-Att [37]	0.0136	0.94	36.91	0.9625
	Swin [48]	0.0135	0.92	36.98	0.9629
	MDTA [39]	0.0133	0.92	37.05	0.9628
	Linear SRA [49]	0.0133	0.93	36.95	0.9629
	EA [50]	0.0135	0.91	36.90	0.9627
	FT-Attention [4]	0.0133	0.91	37.02	**0.9630**
	Ours	**0.0132**	**0.89**	**37.14**	**0.9630**
RICE2	RW-Att [37]	0.0150	1.16	36.56	0.9187
	Swin [48]	0.0149	**1.15**	36.49	0.9186
	MDTA [39]	0.0151	1.17	36.56	0.9183
	Linear SRA [49]	0.0150	1.18	36.60	0.9181
	EA [50]	0.0155	1.20	36.43	0.9183
	FT-Attention [4]	0.0150	1.18	36.64	0.9186
	Ours	**0.0148**	**1.15**	**36.76**	**0.9189**

This confirms the dictionary learns an accurate, disentangled basis for real-world land cover.

This dual evidence links dictionary accuracy to performance: specialized atoms act as priors to infer cloud-occluded content, avoiding generic reconstructions, reducing bias, preserving details, and enhancing robustness. Thus, the CADict-Att mechanism's accurate, discriminative dictionary is key to superior cloud removal.

4.3.3. Comparison with other attention mechanisms

To validate the efficacy of CADict-Att, we conducted comparative experiments by replacing CADict-Att with six efficient attention mechanisms in our network, with results on RICE1 and RICE2 summarized in Table 6.

Standard full-attention, with quadratic computational complexity, is impractical for high-resolution cloud removal. Thus, we focused on efficient variants widely adopted in the field: RW-Att [37] (backbone) captures regional dependencies via rectangular windows but lacks global context. Swin Attention [48] emphasizes local dependencies through shifted windows. MDTA [39] models global channel-wise relationships but neglects spatial details. Linear SRA [49] reduces tokens via pooling for global modeling, sacrificing pixel-level precision. Efficient Attention (EA) [50] and FT-Attention [4] approximate softmax attention with linear complexity to capture global pixel relationships.

While these mechanisms can capture local or global dependencies within a single image, they do not account for potential correlations among samples, limiting their capacity for comprehensive feature extraction. In contrast, CADict-Att captures cross-image correlations, offering a holistic and content-aware feature representation. As demonstrated by the results in Table 6, CADict-Att consistently outperforms all other attention mechanisms across multiple evaluation metrics. This highlights its effectiveness in modeling latent cross-sample correlations, ultimately producing higher-quality cloud-free images.

4.3.4. Ablation study on transformer block numbers

To investigate the impact of Transformer block count on performance and practicality, we designed three model variants (Tiny, Small, Big) with distinct block configurations: [1,1,4,1,1], [1,2,8,2,1], and [2,4,10,4,2]. Comparative experiments were conducted on the RICE1 and RICE2 datasets, with Tiny/Small trained for 1000 epochs and Big for 1500 epochs (to accommodate its larger capacity). Results are summarized in Table 7.

On RICE1, the Big variant achieves the optimal performance (MAE = 0.0131, PSNR = 37.22, SSIM = 0.9635). In contrast, on RICE2, the Small variant delivers the best comprehensive performance (MAE = 0.0148, SAM = 1.15, PSNR = 36.76), while the Big variant exhibits marginal degradation (MAE = 0.0153, PSNR = 36.62) due to potential over-parameterization—its expanded capacity exceeds the demand of small-to-medium datasets, requiring more training data to converge stably.

The Small variant ([1,2,8,2,1]) is selected as the optimal configuration, balancing superior restoration metrics, computational efficiency (50.85G FLOPs), and parameter parsimony (5.93M)—critical for practical remote sensing deployment.

4.3.5. Ablation study on hyperparameter M

To determine the optimal number of atoms (M) in the CADict-Att module, we conducted an ablation study following strict methodological protocols: models were trained on the training set and evaluated on a dedicated validation set to avoid data leakage. The results are detailed in Table 8. The analysis indicates that $M = 128$ achieves the optimal performance across both optical (RICE1/RICE2) and multi-modal (SEN12MS-CR) datasets. While increasing M raises computational complexity, it yields diminishing returns and can even degrade performance on complex multi-modal data due to spectral redundancy. Consequently, $M = 128$ is selected as the default value for all subsequent experiments to balance performance and efficiency.

5. Conclusion

In this research, we presented DictCR-former, a dictionary-based transformer architecture designed to overcome the informational scarcity inherent in single-image cloud removal. By integrating a Content-Aware Dictionary Attention (CADict-Att) mechanism, the framework effectively captures cross-image correlations through learned semantic atoms, allowing for the adaptive reconstruction of obscured regions while maintaining high spatial consistency. Rigorous evaluations across four benchmark datasets demonstrate that DictCR-former significantly outperforms state-of-the-art methods in restoration

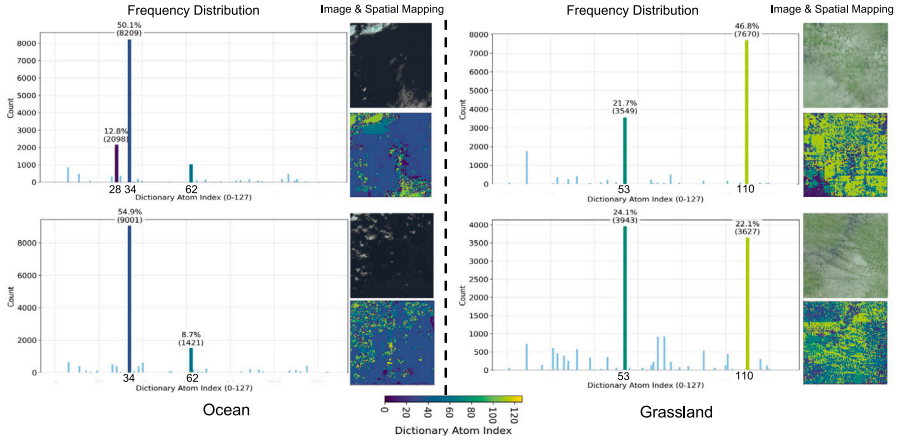


Fig. 9. Quantitative analysis of dictionary atoms (ocean and grassland scenes): Frequency distribution and spatial mapping of top-1 attention scores, demonstrating tight clustering of key categories around specific atoms (ocean: ~34, grassland: ~53, clouds: ~110).

Table 7
Ablation study on the number of Transformer blocks. Metrics with ↓ indicate lower is better, and ↑ indicate higher is better. The best results are in bold.

Dataset	Model	Block numbers	MAE ↓	SAM ↓	PSNR ↑	SSIM ↑	Params(M)	FLOPs(G)
RICE1	Tiny	[1,1,4,1,1]	0.0137	0.92	36.76	0.9626	3.44	36.36
	Small	[1,2,8,2,1]	0.0132	**0.89**	37.14	0.9630	5.93	50.85
	Big	[2,4,10,4,2]	**0.0131**	0.90	**37.22**	**0.9635**	7.78	80.65
RICE2	Tiny	[1,1,4,1,1]	0.0149	1.16	36.67	0.9180	3.44	36.36
	Small	[1,2,8,2,1]	**0.0148**	**1.15**	**36.76**	0.9189	5.93	50.85
	Big	[2,4,10,4,2]	0.0153	1.19	36.62	**0.9192**	7.78	80.65

Table 8
Ablation study on hyperparameter M (number of atoms in CADict-Att) evaluated on validation sets. Metrics with ↓ indicate lower is better, and ↑ indicate higher is better. Best results are in bold.

Datasets	Parameter M	MAE↓	SAM↓	PSNR↑	SSIM↑	Param(M)	FLOPs(G)
RICE1	M = 64	0.01281	0.848	37.129	0.9598	5.87	49.2
	M = **128**	**0.01263**	**0.831**	**37.260**	**0.9604**	5.93	50.8
	M = 256	0.01291	0.839	37.180	0.9603	6.05	54.2
	M = 512	0.01275	0.845	37.222	0.9604	6.30	60.8
RICE2	M = 64	0.01671	1.303	36.020	0.9029	5.87	49.2
	M = **128**	**0.01660**	**1.238**	**36.219**	**0.9034**	5.93	50.8
	M = 256	0.01689	1.274	36.178	0.9027	6.05	54.2
	M = 512	0.01654	1.265	36.152	0.9032	6.30	60.8
SEN12MS-CR	M = 64	0.02353	6.433	29.955	0.9013	5.87	49.2
	M = 128	**0.02296**	**6.411**	**30.166**	**0.9016**	5.93	50.8
	M = 256	0.02382	6.665	29.814	0.9010	6.05	54.2
	M = 512	0.02355	6.537	29.927	0.9006	6.30	60.8

fidelity and computational tractability. We believe this dictionary-driven paradigm offers a valuable new perspective for the remote sensing community, providing an efficient and scalable framework that can be adapted for other severe weather degradation tasks.

Despite these advancements, the model's performance remains constrained by the static nature of the dictionary atoms and a reliance on contextual estimation when thick cloud cover results in absolute signal attenuation. Consequently, future work will focus on dynamic online dictionary learning to improve regional adaptation and the integration of multi-modal data—specifically SAR—to ground reconstructions in physical constraints and enhance spectral precision in restored remote sensing products.

CRediT authorship contribution statement

Wenli Huang: Writing – original draft, Methodology, Funding acquisition, Conceptualization. **Yang Wu:** Methodology, Investigation, Formal analysis. **Sanping Zhou:** Supervision, Project administration, Conceptualization. **Xiaomeng Xin:** Methodology, Investigation, Data curation. **Xiaobo Jia:** Visualization, Validation. **Ping Yu:** Supervision, Resources. **Ye Deng:** Writing – review & editing, Funding acquisition, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

The author(s) declare that no generative AI or AI-assisted technologies were used in the manuscript preparation process, including the generation of **text** and **images**.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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